$\boxed{06.04.2023}$ $x_0, x_1, \dots, x_m \in [a,b]$ $x_i \neq x_j$

$\ell(x) = (x-x_0)(x-x_1)\dots(x-x_m)$ → POLINOM NODAL

$\ell(x_i) = 0$ $\forall\, i=\overline{0,m}$

$\deg(\ell) = m+1$ [grad ℓ ?!]

$P \in \Pi$ (polinom) Π - mulțimea tuturor polinoamelor

Π_m - -||- polinoamelor cu grad $\leq m$

$\Pi_m \subset \Pi_{m+1}$ $\Pi = \bigcup_{m=1}^{\infty} \Pi_m$ Π_m / P_m

$P_m(x) = a_m x^m + a_{m-1}x^{m-1} + \dots + a_1 x + a_0 \in \Pi_m$ $a_m \neq 0$

Restul împărțirii lui P la ℓ (polinomul nodal)?

Teorema împărțirii cu rest pt polinoame

$f, g \in \Pi, g \neq 0 \Rightarrow \exists!\, c, r \in \Pi$ a.î.

$\begin{cases} f = g \cdot c + r \\ \text{grad } r < \text{grad } c \end{cases}$

$\boxed{\text{a) Restul lui } P \text{ la } (X-a)(X-b) = \\ = X^2 - (a+b)X + ab}$

$\ell(x) = (X-x_0)(X-x_1)\dots(X-x_m)$

Obs!! grad $P \geq$ grad $\ell = m+1$

$P = Q \cdot \ell + R$

!! grad $R <$ grad $\ell = m+1 \Rightarrow$ grad $R \leq m$

$P(x) = Q(x)(x-x_0)(x-x_1)\dots(x-x_m) + R(x)$

$x := x_i,\ i=\overline{0,m} \Rightarrow P(x_i) = Q(x_i)\underbrace{\ell(x_i)}_{=0} + R(x_i)$

$P(x_i) = R(x_i)$ $i=\overline{0,m}$

$\left.\begin{array}{l} P(x_i) = R(x_i),\ i=\overline{0,m} \\ \text{grad } R \leq m \end{array}\right\} \Rightarrow R =$ polinomul de interpolare Lagrange pt perechile $(x_i, P(x_i)),\ i=\overline{0,m}$

$f = P$

P.I.L.

$(x_i, f(x_i))$ $i=\overline{0,m}$

? $\exists!\, L$ a.î. grad $L \leq m$ și $L(x_i) = f(x_i)$, $i=\overline{0,m}$

Răspuns: DA $L = L_m(x_0, x_1, \dots, x_m; f)$

Obs!! $f \in \Pi_m \Rightarrow L_m(x_0, \dots, x_m; f) = f$

$f, g \in \Pi_m$ $f \equiv g \Leftrightarrow f(x_i) = g(x_i)$ $i = \overline{1, m+1}$

$\boxed{\text{Interpolare Hermite}}$ → "intră în joc și derivatele"

Să se determine polinomul de grad minim a.î.

$P(a) = f(a);\ P'(a) = f'(a);\ P(b) = f(b)$

Obs!! Un polinom $P \in \Pi_m$ este caracterizat de $m+1$ coeficienți $a_m, a_{m-1}, \dots, a_1, a_0$; pt a-i determina avem nevoie de $(m+1)$ condiții / restricții / ecuații

3 condiții ⇒ 3 coeficienți ⇒ grad $P = 2$ $P(x) = \alpha x^2 + \beta x + \gamma$

$P(x) = \ell(x)\underbrace{f(a)}_{\in\mathbb{R}} + h(x)\underbrace{f'(a)}_{\in\mathbb{R}} + g(x)\underbrace{f(b)}_{\in\mathbb{R}}$

unde $\ell, h, g \in \Pi_2$ care vor fi determinate.

$\begin{cases} \ell(a) = 1 \\ \ell'(a) = 0 \\ \ell(b) = 0 \end{cases}$ $\begin{cases} h(a) = 0 \\ h'(a) = 1 \\ h(b) = 0 \end{cases}$ $\begin{cases} g(a) = 0 \\ g'(a) = 0 \\ g(b) = 1 \end{cases}$

$h(x) = \alpha_2(x-a)(x-b)$

$\left.\begin{array}{l} h'(x) = \alpha_2(x-b+x-a) \\ h'(a) = 1 \end{array}\right\} \Rightarrow \alpha_2(a-b+a-a) = 1 \Rightarrow \alpha_2 = \frac{1}{a-b}$

$h(x) = \frac{1}{a-b}(x-a)(x-b)$

$g(x) = \alpha_3(x-a)^2$

$g'(x) = 2\alpha_3(x-a)$ $g'(a) = 0$

$g(b) = 1 \Rightarrow \alpha_3(b-a)^2 = 1 \Rightarrow \alpha_3 = \frac{1}{(b-a)^2}$

$g(x) = \frac{1}{(b-a)^2}(x-a)^2$

$\ell(a) = 1,\ \ell'(a) = 0,\ \ell(b) = 0$

$\ell(x) = (x-b)(\alpha_1 x + \beta_1)$

$\ell'(x) = (\alpha_1 x + \beta_1) + \alpha_1(x-b) = 2\alpha_1 x + \beta_1 - \alpha_1 b$

$\begin{cases} \ell(a) = 1 \\ \ell'(a) = 0 \end{cases} \Rightarrow \begin{cases} (a-b)(\alpha_1 a + \beta_1) = 1 \\ 2\alpha_1 a + \beta_1 - \alpha_1 b = 0 \end{cases} \Rightarrow \begin{cases} \alpha_1 a^2 - \alpha_1 ab + \beta_1 a - b\beta_1 = 1 \\ 2\alpha_1 a + \beta_1 - \alpha_1 b = 0 \end{cases}$

...

$\begin{cases} \beta_1 = (b-2a)\alpha_1 \\ \alpha_1 a^2 - \cancel{\alpha_1 ab} + \cancel{ab}\alpha_1 - 2\alpha_1 a - \alpha_1 b = 1 \end{cases}$ $\alpha_1 = \frac{1}{a^2 - 2a - b}$

$\beta_1 = \dots$

SAU

$P(x) = \alpha x^2 + \beta x + \gamma$; $P'(x) = 2\alpha x + \beta$

$P(a) = f(a)$

$P(b) = f(b)$

$P'(a) = f'(a)$

$\begin{cases} \alpha a^2 + \beta a + \gamma = f(a) \\ 2\alpha a + \beta = f'(a) \\ \alpha b^2 + \beta b + \gamma = f(b) \end{cases} \Rightarrow \alpha, \beta, \gamma$

Ex. liste $\begin{cases} P^{(k)}(a) = f^{(k)}(a),\ k = \overline{0,m} \\ P(b) = f(b) \\ P'(b) = f'(b) \end{cases}$ → $\begin{cases} P(a) = f(a) \\ P'(a) = f'(a) \\ \vdots \\ P^{(m)}(a) = f^{(m)}(a) \end{cases}$

$m+3$ condiții ⇒ $m+3$ coeficienți ⇒ $P \in \Pi_{m+2}$

$P_{m+2}(x) = \alpha_{m+2}x^{m+2} + \alpha_{m+1}x^{m+1} + \dots + \alpha_1 x + \alpha_0$ → ?! :)

sau $P_{m+2}(x) = \ell_0(x)f(a) + \ell_1(x)f'(a) + \ell_2(x)f''(a) + \dots + \ell_m(x)f^{(m)}(a) + h_0(x)f(b) + h_1(x)f'(b)$

$\deg(\ell_i) = m+2$; $i = \overline{0,m}$ $\deg(h_j) = m+2$ $j = \overline{0,1}$

$\begin{cases} \ell_0(a) = 1 \\ \ell_0'(a) = 0 \\ \vdots \\ \ell_0^{(m)}(a) = 0 \\ \ell_0(b) = 0 \\ \ell_0'(b) = 0 \end{cases}$ $\begin{cases} \ell_1(a) = 0 \\ \ell_1'(a) = 1 \\ \vdots \\ \ell_1^{(m)}(a) = 0 \\ \ell_1(b) = 0 \\ \ell_1'(b) = 0 \end{cases}$ ... $\begin{cases} h_0(a) = 0 \\ h_0'(a) = 0 \\ \vdots \\ h_0^{(m)}(a) = 0 \\ h_0(b) = 1 \\ h_0'(b) = 0 \end{cases}$ $\begin{cases} h_1(a) = 0 \\ h_1'(a) = 0 \\ \vdots \\ h_1^{(m)}(a) = 0 \\ h_1(b) = 0 \\ h_1'(b) = 0 \end{cases}$

$\ell_0(x) = (x-b)^2 \cdot Q(x)$

$\deg(Q) = m$

?? !

se folosește dezvoltarea în serie Taylor pt Q

$h_1(x) = (x-a)^{m+1}(x-b) \cdot \alpha_1$

$h_1'(b) = 1 \Rightarrow \alpha_1 = \dots$

$\boxed{((x-a)^m)^{(m)} = m!}$

$h_0(x) = \underbrace{(x-a)^{m+1}(\alpha x + \beta)}_{\deg(h_0) = m+2}$; $h_0'(x) = (m+1)(x-a)^m(\alpha x + \beta) + (x-a)^{m+1} \cdot \alpha$

$\begin{cases} h_0(b) = 1 \\ h_0'(b) = 0 \end{cases}$ → $\begin{cases} (b-a)^{m+1}(\alpha a + \beta) = \\ (m+1)(b-a)^m(\alpha b + \beta) + \alpha(b-a)^{m+1} = 0 \end{cases}$

$\int_a^b f(x)\,dx = A_0 f(x_0) + A_1 f(x_1) + \dots + A_m f(x_m) + R(f)$

$x_0, x_1, \dots, x_m \in [a,b]$

$x_i = ?$ $A_i = ?$ a.î. să fie cât mai "competitivă"

Grad de exactitate "r" dacă $\forall\, P \in \Pi_r$ $R(P) = 0$

$\int_a^b P(x)\,dx = A_0 P(x_0) + \dots + A_m P(x_m)$

Lista 3

$\int_a^b f(x)\,dx = \frac{b-a}{6}f(a) + \frac{4(b-a)}{6}f\left(\frac{a+b}{2}\right) + \frac{b-a}{6}f(b) + R(f)$

$x_0 = a,$ $x_1 = \frac{a+b}{2};$ $x_2 = b$

$A_0 = \frac{b-a}{6}$ $A_1 = \frac{4(b-a)}{6},$ $A_2 = \frac{b-a}{6}$

$\int_a^b \alpha f + \beta g = \alpha\int_a^b f + \beta\int_a^b g$

Este suficient să verificăm pt $1, x, x^2, x^3, \dots, x^r$

$1, x, x^2, x^3$ dar nu și pt x^4

$f = 1$ $\int_a^b f = \int_a^b 1 = b-a$

$\frac{b-a}{6}\underbrace{f(a)}_{=1} + \frac{4(b-a)}{6}\underbrace{f\left(\frac{a+b}{2}\right)}_{=1} + \frac{b-a}{6}\underbrace{f(b)}_{=1} = \frac{b-a}{6} + \frac{4(b-a)}{6} + \frac{b-a}{6} = b-a$

$f = x$ ...